Maharaj L M R

maharajlmr.official@gmail.com | LinkedIn | GitHub

Education:

B.E Mechatronics Engineering **CGPA - 7.7**
Chennai Institute of Technology, Chennai, Tamil Nadu (Batch 2024-2028)

Projects:

1. **Automated Guided Vehicle (AGV) for Warehouse Automation**
   Developed an AGV prototype using embedded systems and ROS for automated warehouse transport. Implemented sensor-based obstacle detection and motor control to enable safe material movement and improve logistics efficiency.
2. **Agricultural Multi-Function Robot**
   Designed an autonomous agricultural robot capable of seeding, ploughing, and soil preparation. Modeled in SolidWorks and implemented sensor-based navigation with embedded control systems.
3. **AI-Based Defective Package Sorting Conveyor for United Parcel Service (UPS)**
   Built a computer vision-based conveyor system to detect and sort defective packages. Used image processing and a servo-actuated deflection mechanism; prototype demonstrated to UPS engineers.
4. **Magnetic Wall-Climbing Rover**
   Developed a magnetic wall-climbing rover for rust cleaning and inspection of ship hulls, using magnetic adhesion and a motorized drive system for stable movement on vertical metal surfaces.
5. **Hand Gesture Controlled Robotic Arm**
   Developed a robotic hand that mimics human movements using OpenCV-based real-time gesture recognition. Translated webcam-detected hand gestures into robotic arm actions, enabling safe remote operation in hazardous environments such as bomb disposal scenarios.
6. **Low Cost Automation for CNC Loading and Unloading**
   Developed a robotic arm system from scratch for CNC lathe automation, covering design, embedded software, assembly, and manufacturing, and implemented an intuitive and efficient teaching interface to reduce dependence on expensive teach pendants.

Skills:

1. **CAD & Design Tools:** SolidWorks, CATIA, AutoCAD, Siemens NX CAD, 3D Assembly Modeling, Mechanism Design
2. **Analysis & Simulation:** ANSYS (Finite Element Analysis – FEA), Basic Kinematic and Dynamic Analysis
3. **Programming Languages:** C, C++, Python
4. **Robotics & Computer Vision:** ROS 2, OpenCV, Gazebo, YOLOv8, CNN
5. **Embedded Systems:** Raspberry Pi, ESP32, Arduino, Jetsson Nano
6. **Hardware & Fabrication:** Mechanical Assembly, Prototyping, Soldering, Testing & Troubleshooting
7. **Industrial Robot Operation**: Trained in operating the MELFA RV-4F-LD, including programming, teach operations, and safe handling, with practical experience in RT ToolBox for robot control and simulation.

Internship Experience:

1. **Lakshmi Engineering** **(May 2025 - June 2025)**
   Completed a 2-month internship focused on industrial manufacturing and production systems. Assisted in machining, mechanical assembly, and fabrication of precision components while contributing to quality inspection. Gained practical exposure to CNC machining, maintenance practices, and industrial production workflows.
2. **Chennai Port Authority** **(Nov 3, 2025 – Nov 29, 2025)**
   Completed training in the Mechanical and Electronics departments, studying Caterpillar 3412 locomotive diesel engines, traction systems, and cooling systems. Gained exposure to Battery Management Systems (BMS) and locomotive power conversion from mechanical to electrical energy for traction motors.
3. **National Institute of Technology Calicut (NIT Calicut)** **(May 2026 – Sept 2026)**
   Completed a 4-month research internship in Robotics and Artificial Intelligence under **Dr. Shihabudheen K V**, Developed AI-powered robotic applications using the **Hiwonder JetAcker AI Robot Kit** with **NVIDIA Jetson**, working extensively with **ROS 1, ROS 2, Gazebo, SLAM, computer vision, voice control, and multimodal AI (ChatGPT)**. Gained hands-on experience in autonomous navigation, AI vision interaction, sensor integration, and robotic software development.

Achievements:

1. **2nd Runner Up – Bonfiglioli ×CIT 48-Hour Hackathon 2026**
   Won ₹1,00,000 prize for developing an Automated Guided Vehicle for warehouse automation, competing among 1000+ teams.
2. **1st Runner Up – DriveNext RANE SLD Hackathon 2026**
   Won ₹75,000 prize for developing a 5-DOF robotic arm for low-cost CNC lathe loading and unloading automation.
3. **Selected for Smart India Hackathon (SIH 2025) – Institutional Round Selection**
   Selected for presenting the Agricultural Multi-Function Robot project at the institutional round screening.

Certifications (Courses & Competition):

1. Entrepreneurship Hackathon – STPI x Techniche
2. SmartMotion Hackathon 2.0
3. Introduction to Internet of Things – NPTEL
4. Developing Soft Skills & Personality – NPTEL
5. CCNA: Introduction to Networks (Cisco)
6. AutoCAD Completion Certificate
7. SolidWorks Completion Certificate – CADD Centre